Exact normalized scaling and full closure under a target-row change

2026-11-14. Source `85c57a726a644a6d2138a5c97785ab1119a06aff`. All claims unsigned [pcc:unverified].

The normalized target row has an exact inverse-pair pole-scaling law. More decisively, the actual change of rows preserves the full iterated degree-budgeted closure whenever the budget admits both input families. This is a statement about the actual symbolic coefficient algebra, not just equality of finite-field solution sets. It isolates what this normalization can change and redirects the search for a closure improvement beyond a constant row basis [pcc:unverified].

Central claims: normalized scaling `PCC-20261114-46dee9`, generic full closure `PCC-20261114-5f67e7`, actual normalized row `PCC-20261114-044d98`, actual full closure `PCC-20261114-d07690`. Gate/build receipt `PCC-20261114-f6eb11`. Scaling source, closure source, prospective contract, exact proof map, editable diagram, SVG, PDF.

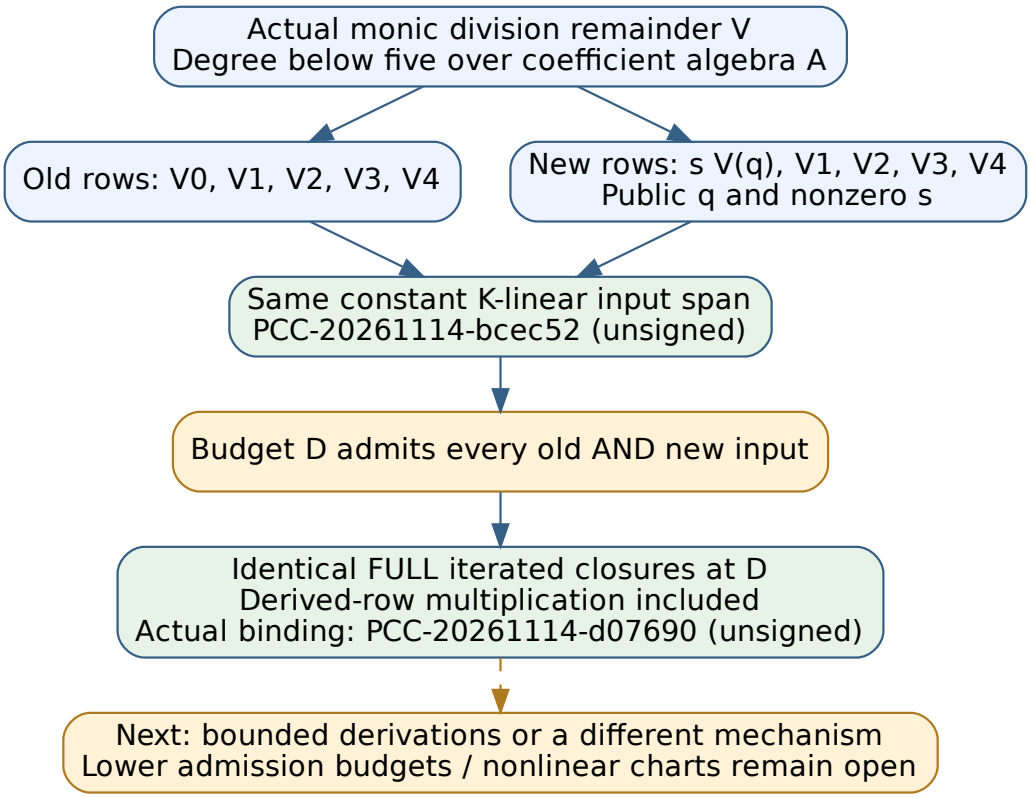


Figure 1: The exact row transformation preserves the full closure once both sets of inputs are admitted. The lower-budget and nonlinear routes remain separate. Evidence unsigned.

Exact normalized scaling

Keep the actual monic norm quotient, actual Dickson dividend and actual membership quotient from the previous component. Write $d = 2^{n+1}$ and R_e for the membership remainder. Its normalized target row is

$$T_e = e^{d-3}(M_n(2)(r) - c) - (c_0 + c_1 e^2 r) J_e Q_e(e^2 r),$$

where Q_e is the computed monic quotient and

$$J_e = 2za_1 + 4e^2 zra_2 + 6e^4 zr^2 - e(3r^2 + a)(c_0 + c_1 e^2 r) - 2e^3 z^2 c_1.$$

The prior actual division gives $R_e(e^2 r) = e^3 T_e$. Parameter one is exactly $T_1 = R_1(r)$. Evaluating the full-remainder transport at the scaled target and cancelling only the explicit nonzero factor gives

$$e^{d-3} T_1(a_1 l^4, a_2 l^2, c_0 l^3, c_1 l) = T_e(a_1, a_2, c_0, c_1), \qquad el = 1.$$

The curve premise $z^2 = r^3 + ar + b$ and $d \geq 3$ remain explicit. The identity holds at every stated depth over a field; it is not yet an identification with a formal multivariate weighted homogeneous component. Thus weight29 at depth4 remains a candidate upper bound, not an established exact degree [pcc:unverified].

Full closure, rather than static input multiples

Let A be a commutative algebra over a field K, and let $\delta : A \to \mathbb{N}$ be an explicit degree function. The inductive closure at budget D allows:

1. an input f only when $\delta(f) \leq D$;
2. zero, addition and multiplication by a scalar in K;
3. multiplication of **any already derived row** f by $a \in A$ when $\delta(a) + \delta(f) \leq D$.

These are the literal full-closure rules of the existing ML source, rebuilt at this study's pinned toolchain. Ordinary total degree and weighted total degree can each instantiate the explicit function. No static Macaulay-space surrogate, available GB or normal-form oracle is assumed [pcc:unverified].

If input sets F, G have equal K-linear spans and every member of both sets is admitted at D, their complete closures at D coincide. The proof first puts each input span in the opposite closure, then transports every derivation, including multiplication of derived rows. It needs no theorem about the size of a solution set [pcc:unverified].

Binding to the actual normalized target row

For an actual degree-below-five remainder $V \in A[X]$, public $q \in K$ and public $s \neq 0$, the old inputs are its five coefficients. The new inputs are $sV(q)$ and coefficients1 through4. Polynomial evaluation gives

$$V(q) = V_0 + qV_1 + q^2 V_2 + q^3 V_3 + q^4 V_4.$$

This is an invertible constant row transformation: solve for V_0 using s^{-1}. Hence the generator spans agree. For the actual membership remainder the degree guard follows from monic norm/support division; $q = e^2 r$ and $s = (e^3)^{-1}$ give exactly T_e. Public curve and parameter values are mapped into A by its algebra map, while the retained coefficients remain arbitrary elements of A. In particular, A may be the multivariate polynomial ring in those coefficients: this is a symbolic identity, not equality after sampling [pcc:unverified].

Consequently, once both original and normalized input families fit D, the same rows are derivable in their full closures. The theorem keeps both admission hypotheses explicit; it does not assert a numerical admission threshold for an unproved weighted grading [pcc:unverified].

What the test decides

Constant row recombination alone cannot lower the full closure above that input-admission threshold. This tests the proposed mechanism before using a degree fall as evidence of a faster solver. It leaves open effects below the threshold, implementation changes, different degree functions, nonlinear charts, different factor bases and other non-generic algorithms. It does not show that the original closure is large: the transformed row may still expose short derivations already present in it. It is not a generic-group lower bound or a global impossibility result [pcc:unverified].

The exact transformation and scaling remain useful algebraic information. A subsequent candidate needs to change something beyond this constant input span, or supply actual bounded derivations and a complete solver cost. The original nonscalar-origin/jet family still needs its own faithful map; this norm normalization is not a map to that family [pcc:unverified].

Sources, validation and retained failures

Sources checked (KB): unavailable as previously documented, no unchanged retry. The bounded receipt records three fallback queries and actual primary sources. Read ML commit `8020e131a1a3806531d695b6bc7586473802dada` `formal/ICFallDegrees/{Closure,VariableClosure,FiniteQuotient,PointIdeal}.lean`. The finite-quotient result needs an available GB/reduced-ideal subsystem; the point-ideal result needs budgeted coordinate derivations. Neither discharges those premises from a small zero set. Prior weighted-GB source review and Mathlib's weighted homogeneous definitions remain relevant. This bounded review establishes no novelty or exhaustive absence [pcc:unverified].

Final normalized build002 and closure build005 exit0. There are15 authored theorem bindings. The whole-study gate audits369 declarations, including the new generated `InClosure.brecOn`, and reports0 additional trusted uses, exit0. Standard foundations only. Exact source/log hashes are in the gate receipt, claim `PCC-20261114-f6eb11`; theorem claims are unsigned/default not_covered (exit2). Import-inclusive fresh replay exits0, including both new modules; validation receipt PCC-20261114-6643d4 binds the exact hashes [pcc:unverified].

Retain normalized draft001's parameter-one wrapper error, closure drafts001 through003's predicate, field, notation, recursive parameter and coercion errors, and draft004's unnecessary tactic-sequencing warning. The final source fixes these without changing statements, suppressing checks or adding trusted axioms. No Rust/native mathematical source changed and no prior native test was counted again. The inherited canonical SHA256 clippy and benchrec diagnostics remain untouched [pcc:unverified].

No fixture, unknown-root, F4/msolve, paid/cloud/attack or timing experiment was attempted. The earlier native fixture remains1 planned,0 attempted, 0 completed,0 excluded,1 admission-blocked by checkpoint upload transport. Local Lean proof construction is separate. Complete growing-family closure, an efficient public-input finder, extraction and expected bit cost remain open [pcc:unverified].

Global performance/curve/attack-route graphs are unchanged: this component adds no measured speedup or group-transport edge. All previous sources, evidence, claims and canonical ledger prefix are preserved. The new visual records the exact mechanism and its scope [pcc:unverified].

Exact proof map

All entries unsigned [pcc:unverified]. Declaration/context hashes and source freeze are in claim-pins.json.

Declaration	Claim
`NormTargetGrading.actualTargetFactor`	`PCC-20261114-17b15a`
`NormTargetGrading.normalizedOne`	`PCC-20261114-4f506d`
`NormTargetGrading.membershipOne`	`PCC-20261114-d8ecad`
`NormTargetGrading.targetValueScale`	`PCC-20261114-f41fe3`
`NormTargetGrading.normalizedScale`	`PCC-20261114-46dee9`
`NormRowClosure.transport`	`PCC-20261114-7326e0`
`NormRowClosure.inputSpan`	`PCC-20261114-d0640f`
`NormRowClosure.closureSpanInvariant`	`PCC-20261114-5f67e7`
`NormRowClosure.evaluationRows`	`PCC-20261114-a28623`
`NormRowClosure.rowsMembership`	`PCC-20261114-a2c68b`
`NormRowClosure.rowSpans`	`PCC-20261114-bcec52`
`NormRowClosure.rowClosure`	`PCC-20261114-1f8f28`
`NormRowClosure.actualMembershipWindow`	`PCC-20261114-87bdd8`
`NormRowClosure.actualNormalizedRow`	`PCC-20261114-044d98`
`NormRowClosure.actualRowClosure`	`PCC-20261114-d07690`